Baran Erdogan – Cloud Platforms – Assignment #3

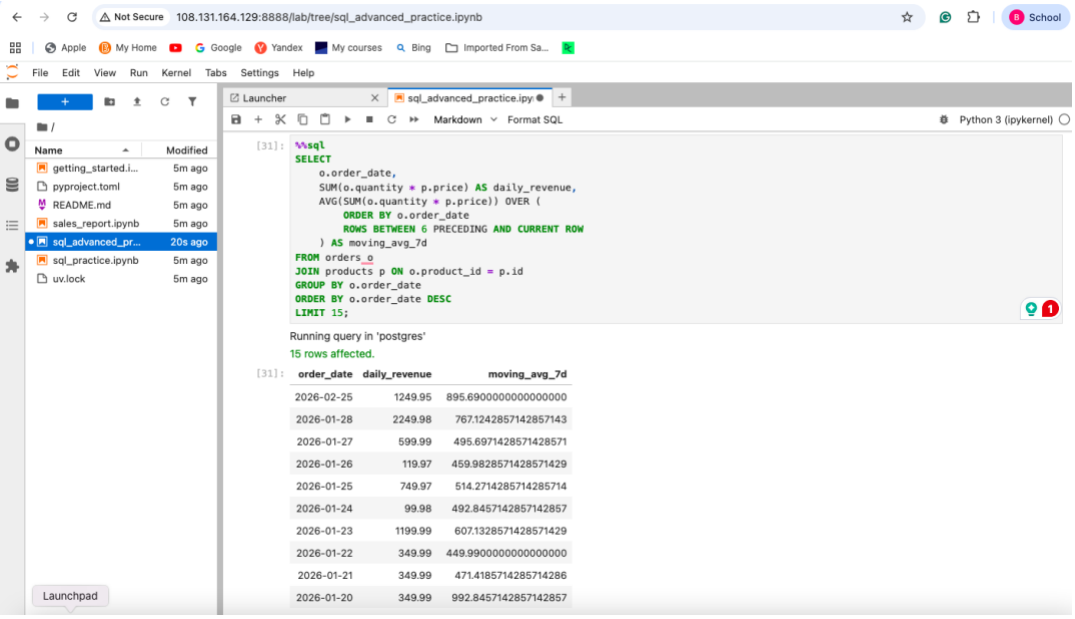

Figure 1: Query #1

The query multiplies quantity from the orders table with price from the products table, to derive the revenue. Applying a 7-day moving average helps identify if sales are trending up or down, filtering out daily noise.

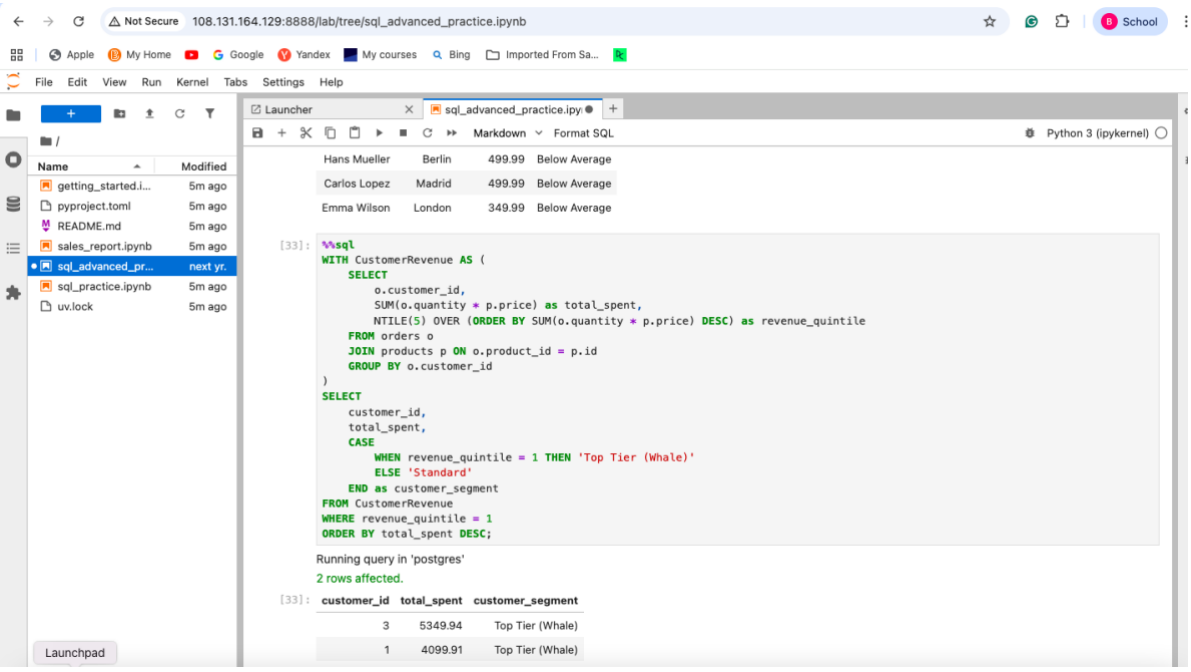

Figure 2: Query #2

This query utilizes a CTE to handle the initial aggregation and a Window Function to split the customer base into five equal groups based on their total spend.

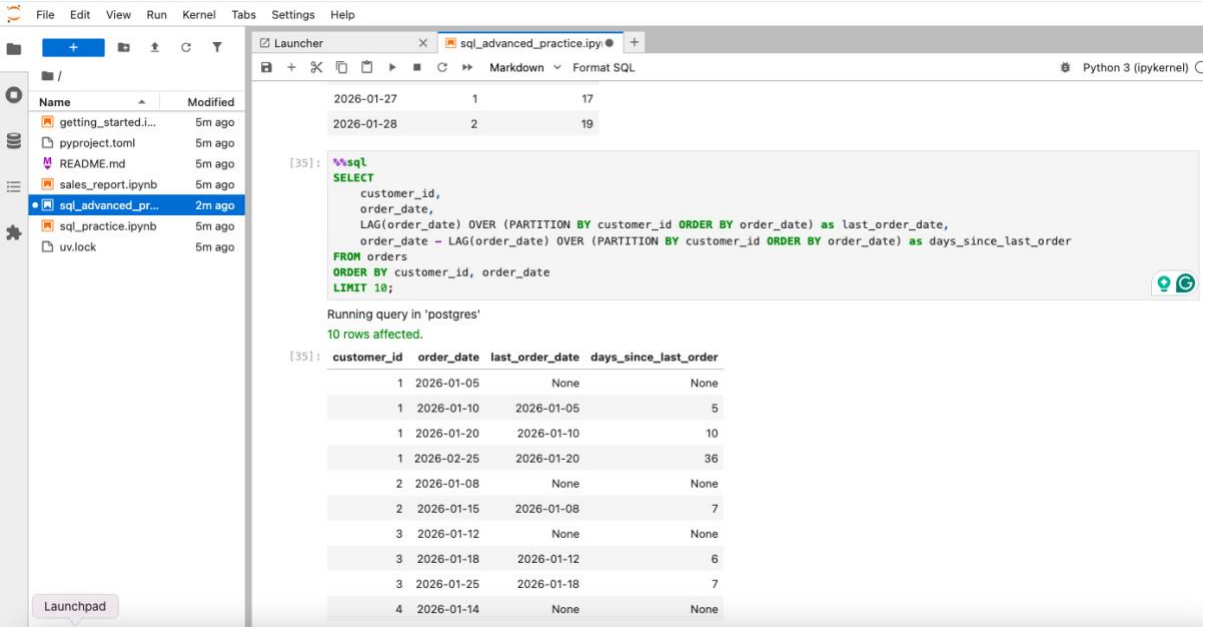

Figure 3: Query #3

This query uses the LAG window function to compare each order's date against the customer's previous purchase to calculate the exact time gap between transactions. By measuring this purchase velocity, a business analyst can identify slowing buying patterns and predict potential customer churn before it happens.

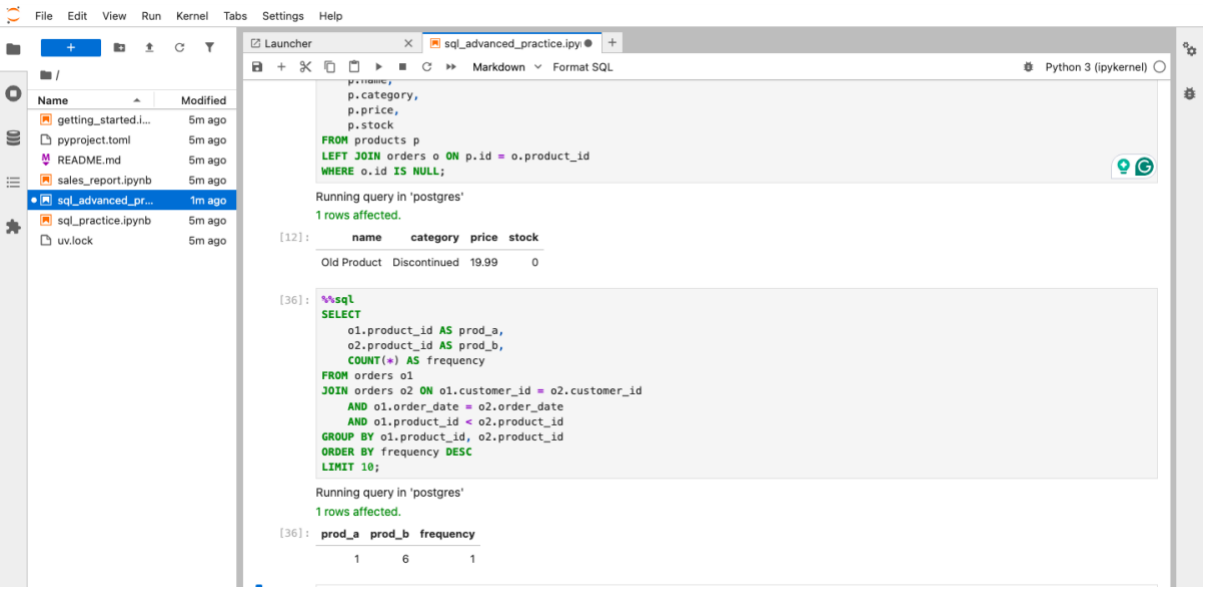

Figure 4: Query #4

This query performs a Self-Join on the orders table to create every possible pair of products purchased by a single customer during a single shopping trip. By counting these occurrences, we can identify high-affinity product pairs to drive “Frequently Bought Together" recommendations and optimize store layout.

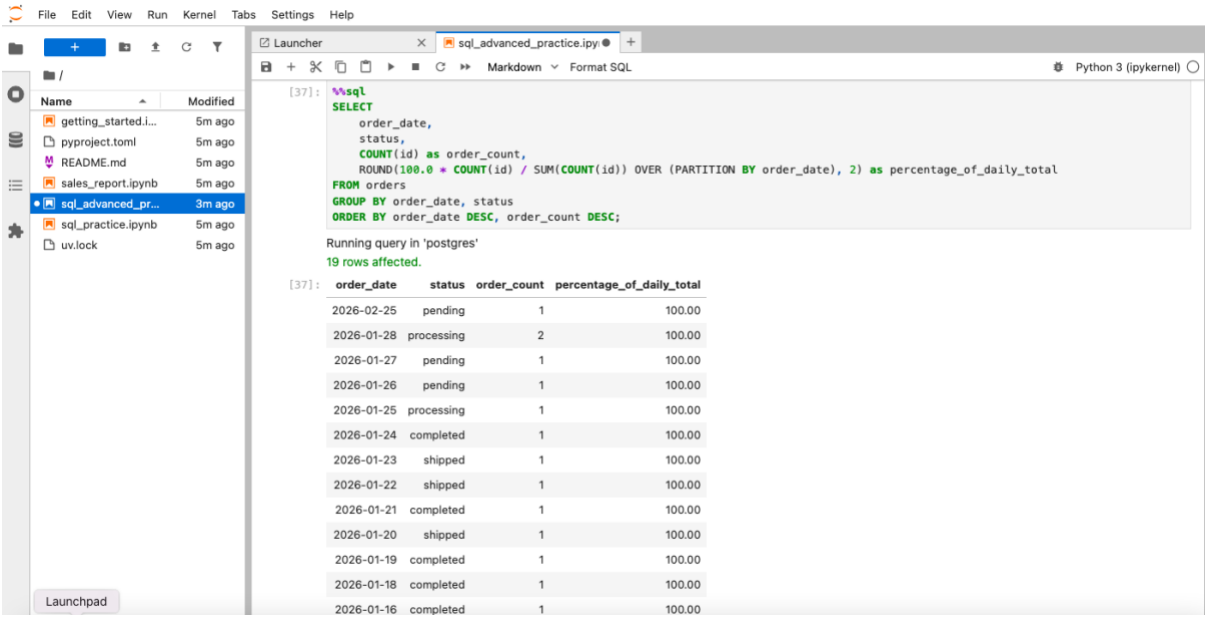

Figure 5: Query #5

This query calculates the daily distribution of order statuses by combining a standard aggregation with a windowed partition to determine each status’s share of the day's total volume.